I. k-means Core Concepts (1–10)

1. What is the fundamental goal of clustering in the context of unsupervised learning?
Answer: To **group similar datapoints together** and separate dissimilar datapoints without using labeled data.

2. How do points within a single cluster compare to each other, and how do points between different clusters compare?
Answer: Points **within a cluster are similar** to each other; points **between clusters are dissimilar**.

3. What is the k-means clustering algorithm designed to do to a dataset?
Answer: Partition the dataset into (k) clusters by **minimizing the sum of squared distances** of points to their cluster centers.

4. In the k-means algorithm, what is the term for the center of a cluster to which all points in that cluster are assigned?
Answer: Cluster mean or **centroid**.

5. Given (n) datapoints, (\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n), what is the goal of k-means clustering in relation to the sum of variances?
Answer: To **minimize the Within-Cluster Sum of Squares (WCSS)**, i.e., the sum of squared distances from points to their cluster centroids.

6. What does the acronym WCSS stand for in the context of k-means clustering?
Answer: Within-Cluster Sum of Squares.

7. Write the formula for the Within-Cluster Sum of Squares (WCSS), where (k) is the number of clusters and (\vec{m}_j) is the mean of cluster (C_j).
Answer:
[
\text{WCSS} = \sum_{j=1}^{k} \sum_{\vec{x}_i \in C_j} |\vec{x}_i - \vec{m}_j|^2
]

8. When a point (\vec{x}_i) is assigned to cluster (C_j), what geometric property must the cluster mean (\vec{m}_j) satisfy in relation to the point?
Answer: (\vec{m}_j) is the **closest centroid** to (\vec{x}_i) among all cluster centroids.

9. In the k-means objective function, what distance metric is typically used to measure the distance between a point (\vec{x}_i) and a cluster mean (\vec{m}_j)?
Answer: Squared Euclidean distance: (|\vec{x}_i - \vec{m}_j|^2).

10. What does the term ($|C_j|$) represent in the k-means algorithm's mean calculation formula?
Answer: The **number of points assigned to cluster (C_j)**.

II. The k-means Algorithm Steps (11–20)

11. What is the first step in the k-means clustering algorithm regarding cluster means initialization?
Answer: Initialize (k) cluster centroids, either randomly or using a heuristic.

12. What are the two main methods used to initialize the cluster means for the first iteration?
Answer:

1. **Random selection** of (k) points as initial centroids.
2. **Random assignment** of points to clusters and then computing initial centroids.

13. If the first method of initialization is used, how many datapoints are chosen to be the initial cluster means?
Answer: Exactly (k) datapoints are selected.

14. If the second initialization method is used, how are the (n) datapoints initially assigned to the (k) clusters?
Answer: Points are **randomly assigned** to clusters, and cluster means are computed from these assignments.

15. What is the process of Assignment (Step 1B) in the k-means algorithm at the (k^{th}) iteration?
Answer: Assign each datapoint to the **nearest cluster centroid** based on distance.

16. Write the mathematical condition for assigning a datapoint ($\vec{x}_i$) to cluster ($C_j^{(k+1)}$) during the assignment step.
Answer:

$$\vec{x}_i \in C_j^{(k+1)} \quad \text{if} \quad |\vec{x}_i - \vec{m}_j^{(k)}|^2 \le |\vec{x}_i - \vec{m}_l^{(k)}|^2 \quad \forall l$$

17. What is the process of Mean Update (Step 2) in the k-means algorithm?
Answer: Recalculate each cluster mean as the **average of all points assigned** to that cluster.

18. Write the formula for calculating the updated cluster mean ($\vec{m}_j^{(k+1)}$) after the Assignment step.
Answer:

$$
\vec{m}_j^{(k+1)} = \frac{1}{|C_j^{(k+1)}|} \sum_{\vec{x}_i \in C_j^{(k+1)}} \vec{x}_i
$$

19. What is the condition for the k-means algorithm to terminate (Check convergence)?
Answer: When **cluster assignments no longer change** or the change in WCSS is below a threshold.

20. If the assignment of points to clusters has changed between two iterations, what must the algorithm do next?
Answer: Recompute the cluster means and repeat the assignment step.

III. Mathematical and Practical Aspects (21–30)

21. What is the minimum possible value for the number of clusters (k)?
Answer: ($k_{\min} = 1$).

22. The term ($k \le n$) must be satisfied for the number of clusters. What does (n) represent?
Answer: The **total number of datapoints**.

23. The squared Euclidean distance between two vectors ($\vec{x}_i$) and ($\vec{m}_j$) is represented by what symbolic expression in the k-means context?
Answer: ($\|\vec{x}_i - \vec{m}_j\|^2$)

24. If the initial choice of cluster means is random, what is a potential issue with the final clustering result?
Answer: The algorithm may converge to a **local minimum**, resulting in **suboptimal clusters**.

25. Which characteristic of the cluster means (that they are closer to the points within their cluster) is the k-means algorithm continually trying to improve?
Answer: Minimization of intra-cluster variance (WCSS).

26. How does the assignment of clusters, ($I(\vec{x}_i \in C_j)$), behave (what values does it take) when ($\vec{x}_i$) is assigned to a cluster?
Answer: It is a **binary indicator**: 1 if ($\vec{x}_i \in C_j$), 0 otherwise.

27. The mean update step essentially computes the centroid of the assigned points. What is the statistical term for this?
Answer: Arithmetic mean (or sample mean).

28. What happens to the cluster mean if a cluster is assigned only one point?
Answer: The cluster mean becomes **exactly that single point**.

29. If the number of clusters ((k)) is not known, what is a common technique used to estimate the optimal value for (k)?
Answer: The **Elbow Method**, which plots WCSS vs. (k) and identifies the "elbow" point.

30. In terms of cluster variance, what is the ultimate goal achieved by minimizing the WCSS?
Answer: To make **points within each cluster as similar as possible**, achieving compact and well-separated clusters.